PCAP Network Analysis

Problem Statement & Objectives

The problem

Given a PCAP file, show which protocols are present, who talks to whom, and how each TCP connection behaves.

Analysis a PCAP file on Wireshark seems difficult.

Our objectives

1. Upload a PCAP file and click Analyze.
2. Show an overview first, details only on click.
3. Explain protocols, flows, RTT, retransmissions, flow control, congestion control.
4. Simplify PCAP file analysis and make it beginner friendly.

Input

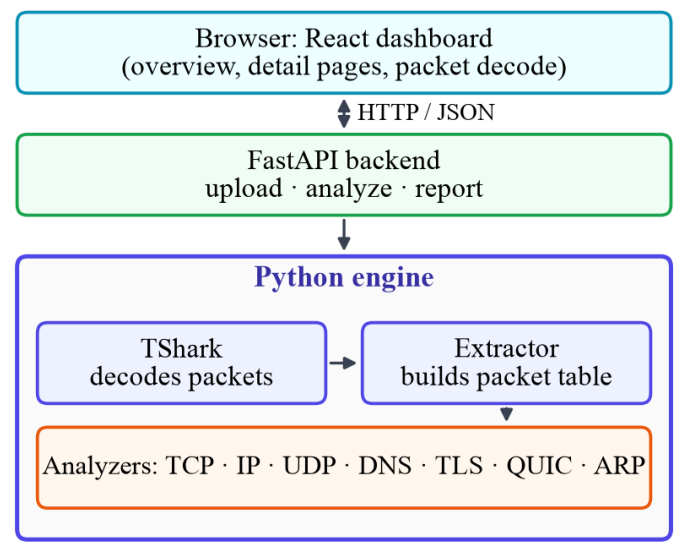
One PCAP or PCAPNG file, uploaded in the browser.

Output

An overview page, detail pages and a JSON report.

Architecture & How It Works

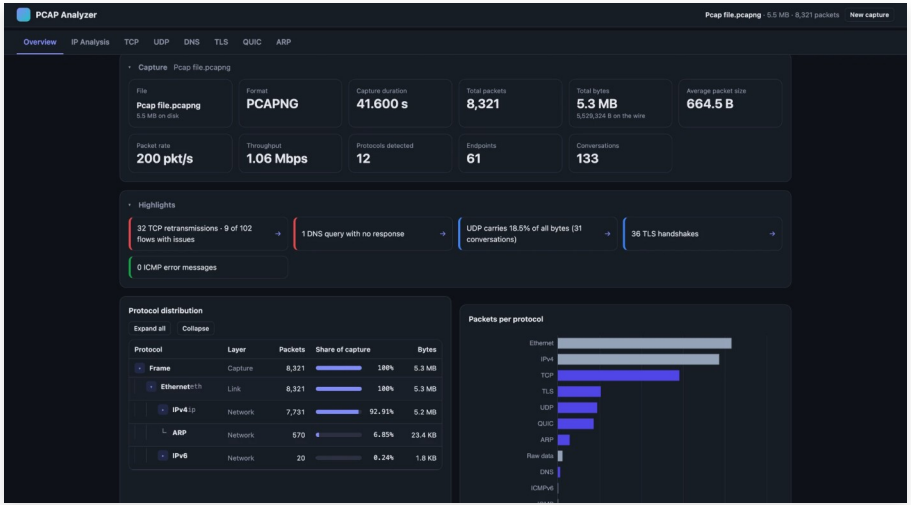
TShark reads the packets. Our Python code adds up the results.



1. **Upload**
   The browser sends the PCAP file to our FastAPI server.
2. **Decode**
   TShark (Wireshark's command-line tool) reads every packet.
3. **Analyze**
   Our Python engine groups packets into flows and does the maths.
4. **Show**
   A React dashboard shows an overview, then detail pages.

Dashboard Flow

The core of the project: from upload to packet details

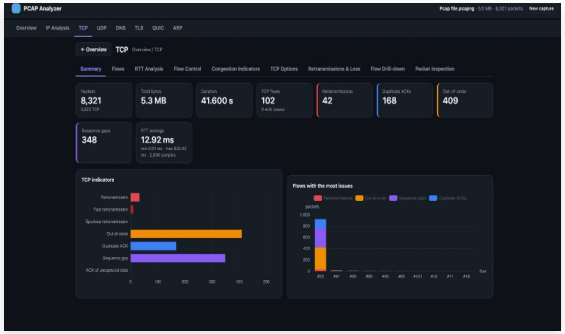


Top navbar on every page: Overview · IP Analysis · TCP · UDP · DNS · TLS · QUIC · ARP

1. **Upload**
   Pick a PCAP file and click Analyze.
2. **Overview**
   Capture facts, highlights, and a chart of all protocols.
3. **Pick a protocol**
   Click a card or the navbar to open its page.
4. **Detail tabs**
   Metrics, charts and tables.
5. **Packet Inspection**
   Click any packet to see its full decode.

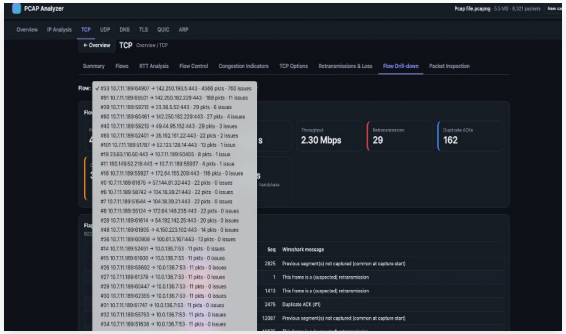
Detail Pages

Detail pages exist for TCP, UDP, DNS, TLS, QUIC, ARP and IP



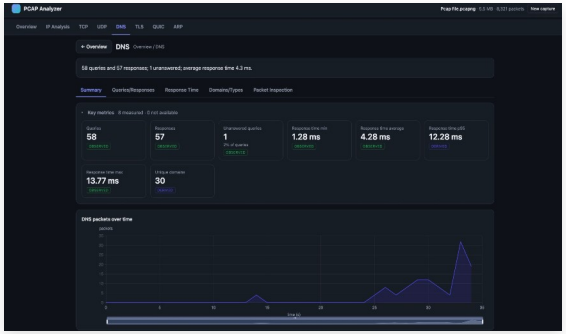
TCP page: 9 tabs

Summary, Flows, RTT, Flow Control, Congestion, Options, Retransmissions, Flow Drill-down, Packet Inspection.



Flow Drill-down

Pick one flow and see its numbers and the Wireshark message for each problem packet.



DNS page: 5 tabs

Queries, response time and domains. Every number is tagged Observed or Derived.

Challenges Faced

And how we solved them

1	**TShark field names change**	Names differ between versions. Our code checks which fields exist first.
2	**Big files use a lot of memory**	We read TShark output in chunks, so memory stays near 4 KB per packet.
3	**Some charts were not reliable**	We removed charts we could not trust instead of showing doubtful numbers.